Final Exam

Graded

Student

Jinghan Zhang

Total Points

40 / 45 pts

Question 1

Overall Grades **40** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 5 pts** Point adjustment

MATH 202-001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Jinghan Zhang	jz8542	Jinghan Zhang

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: Jinghan Zhang.

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	4.5	5	5	5	5	5	0.5	5	40

Problem 1. [5] Find the general solution of the linear system:

$$2x - 3y = 1$$
$$-2y - 3z = 1$$
$$x + 2z = 1$$

$$\begin{bmatrix} 2 & -3 & 0 \\ 0 & -2 & -3 \\ 1 & 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right]$$

$R_1 \leftarrow R_1 - 2R_3 \sim \left[\begin{array}{ccc|c} 0 & -3 & -4 & -1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right]$

$R_1 \leftarrow R_1 - \frac{3}{2}R_2 \sim \left[\begin{array}{ccc|c} 0 & 0 & \frac{1}{2} & -\frac{5}{2} \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right]$

$R_1 \leftarrow 2R_1 \sim \left[\begin{array}{ccc|c} 0 & 0 & 1 & -5 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right]$

$R_2 \leftarrow R_2 + 3R_1 \sim \left[\begin{array}{ccc|c} 0 & 0 & 1 & -5 \\ 0 & -2 & 0 & -14 \\ 1 & 0 & 2 & 1 \end{array}\right]$

$R_2 \leftarrow -\frac{1}{2}R_2 \sim \left[\begin{array}{ccc|c} 0 & 0 & 1 & -5 \\ 0 & 1 & 0 & 7 \\ 1 & 0 & 2 & 1 \end{array}\right]$

$R_3 \leftarrow R_3 - 2R_1 \sim \left[\begin{array}{ccc|c} 0 & 0 & 1 & -5 \\ 0 & 1 & 0 & 7 \\ 1 & 0 & 0 & 11 \end{array}\right]$

$R_1 \leftrightarrow R_3 \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -5 \end{array}\right]$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \\ -5 \end{bmatrix}$$

∴ the general solution is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \\ -5 \end{bmatrix}$$

4.5/5

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

$A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$

$R_1 \leftarrow R_1 - 2R_4 \sim \begin{bmatrix} 0 & -2 & -2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$

$R_1 \leftarrow R_1 + 2R_3 \sim \begin{bmatrix} 0 & 0 & 0 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$

$R_2 \leftarrow \frac{1}{5} R_2 \sim \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$

$R_4 \leftarrow R_4 + R_2 \sim \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & 3 \end{bmatrix}$

$R_4 \leftarrow R_4 - 3R_3 \sim \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$R_2 \leftarrow -R_2 \sim \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$R_1 \leftrightarrow R_2 \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$R_2 \leftrightarrow R_3 \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(1) From the reduced-echelon form, we can see that column 1 and 2 have pivots.
∴ ~~the~~ basis of the column space of A is $\left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

(2) From the reduced-echelon form, we can see that
~~the~~ basis of ~~the~~ row space of A is $\left\{ \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \right\}$.
−0.5

(3) $Ax = 0$

$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$

$z = t$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

∴ a basis of the null space of A is $\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$.

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$\because A\cdot\begin{bmatrix}0\\1\\-1\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}$, $A\cdot\begin{bmatrix}1\\0\\-1\end{bmatrix} = \begin{bmatrix}0\\2\end{bmatrix}$, $A\cdot\begin{bmatrix}0\\0\\-1\end{bmatrix} = \begin{bmatrix}2\\2\end{bmatrix}$

$\therefore A\cdot\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix} = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}$

$A = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}\cdot\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}^{-1}$

Find the inverse of $\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$

$$\left[\begin{array}{ccc|ccc}0&1&0&1&0&0\\1&0&0&0&1&0\\-1&-1&-1&0&0&1\end{array}\right] \xrightarrow{R_3 \leftarrow R_3+R_1+R_2} \left[\begin{array}{ccc|ccc}0&1&0&1&0&0\\1&0&0&0&1&0\\0&0&-1&1&1&1\end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow -R_3} \left[\begin{array}{ccc|ccc}0&1&0&1&0&0\\1&0&0&0&1&0\\0&0&1&-1&-1&-1\end{array}\right]$$

$$\xrightarrow{R_1 \Leftrightarrow R_2} \left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&0&1&-1&-1&-1\end{array}\right]$$

$\therefore \begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}^{-1} = \begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$

$\therefore A = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}\cdot\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$

$= \begin{bmatrix}-2&-1&-2\\0&-1&-2\end{bmatrix}$

$\therefore T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = \begin{bmatrix}-2&-1&-2\\0&-1&-2\end{bmatrix}\cdot\begin{bmatrix}x\\y\\z\end{bmatrix}$,

where $A = \begin{bmatrix}-2&-1&-2\\0&-1&-2\end{bmatrix}$

Problem 4. [5'] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$

$R_3 \leftarrow R_3 + 2R_1$ $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ 0 & 2 & 1 \end{bmatrix}$

$\sim$

$R_3 \leftarrow R_3 - R_2$ $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

$\sim$

$A - \lambda I = \begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}$

$\det(A - \lambda I) = (2-\lambda) \cdot \begin{vmatrix} 1-\lambda & 0 \\ -2 & 1-\lambda \end{vmatrix}$

$= (2-\lambda) \cdot (1-\lambda)^2$

$\lambda_1 = 1, \ \lambda_2 = 1, \ \lambda_3 = 2$

① $\lambda = 1$. $A - \lambda I = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$v = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, geometric multiplicity $= 1$.

$J(\lambda = 1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

② $\lambda = 2$. $A - \lambda I = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{1}{3} \\ 0 & 1 & \frac{1}{6} \\ 0 & 0 & 0 \end{bmatrix}$.

$v = \begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{6} \\ 1 \end{bmatrix}$. geometric multiplicity $= 1$.

$J(\lambda = 2) = (2)$

$\therefore J = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

Problem 5. [5'] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$X = [x \ y \ z]^T$

$f(x,y,z) = X^T \cdot A \cdot X$

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}$$

$$\det(A-\lambda I) = (3-\lambda)\cdot \begin{vmatrix} 3-\lambda & -1 \\ -1 & 4-\lambda \end{vmatrix} + 1 \cdot \begin{vmatrix} 0 & 3-\lambda \\ 1 & -1 \end{vmatrix}$$

$$= (3-\lambda)\cdot(\lambda^2 - 7\lambda + 11) - 1\cdot(3-\lambda)$$

$$= (3-\lambda)\cdot(\lambda^2 - 7\lambda + 10)$$

$$= (3-\lambda)(\lambda-2)(\lambda-5)$$

$\lambda_1 = 2, \ \lambda_2 = 3, \ \lambda_3 = 5, \quad D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

① $\lambda = 2$, $A - \lambda I = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$

$$v_1 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \quad u_1 = \frac{1}{\sqrt{3}} \cdot v_1 = \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$$

② $\lambda = 3$, $A - \lambda I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$$v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad u_2 = \frac{1}{\sqrt{2}} \cdot v_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$$

③ $\lambda = 5$, $A - \lambda I = \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$

$$v_3 = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \quad u_3 = \frac{1}{\sqrt{6}} \cdot v_3 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}$$

$$P = \begin{bmatrix} -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{bmatrix}$$

$X = PU$, $U = [u \ v \ w]^T$

$$f = X^T A X = (PU)^T A (PU)$$
$$= U^T P^T A P U$$
$$= U^T D U$$
$$= 2u^2 + 3v^2 + 5w^2.$$

5/5

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

$$\begin{bmatrix} 0 & 2 & -1 \\ 2 & -1 & 1 \\ -2 & 2 & 0 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right]$$

$R_2 \leftarrow R_2 + R_3$ $\sim$ $\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 0 & 1 & 1 & 0 \\ -2 & 2 & 0 & -8 \end{array}\right]$

$R_1 \leftarrow R_1 - 2R_2$ $\sim$ $\left[\begin{array}{ccc|c} 0 & 0 & -3 & -3 \\ 0 & 1 & 1 & 0 \\ -2 & 2 & 0 & -8 \end{array}\right]$

$R_1 \leftarrow -\frac{1}{3}R_1$ $\sim$ $\left[\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ -2 & 2 & 0 & -8 \end{array}\right]$

$R_3 \leftarrow -\frac{1}{2}R_3$ $\sim$ $\left[\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & -1 & 0 & 4 \end{array}\right]$

$R_2 \leftarrow R_2 - R_1$ $\sim$ $\left[\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & -1 & 0 & 4 \end{array}\right]$

$R_3 \leftarrow R_3 + R_2$ $\sim$ $\left[\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 3 \end{array}\right]$

$R_1 \leftrightarrow R_3$ $\sim$ $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$

$$\therefore \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}.$$

$a=3$, $b=-1$, $c=1$

Problem 7. [5'] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

First, find the orthogonal basis for V.

$$v_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \quad u_1 = v_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$$

$$v_2 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}, \quad u_2 = v_2 - \frac{u_1 \cdot v_2}{\|u_1\|^2} \cdot u_1 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{(\sqrt{2})^2} \cdot \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$$

$$= \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$$

Now, $u_1 \cdot u_2 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix} \cdot \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix} = 0.$

Second, find the orthonormal basis for V.

$$w_1 = \frac{1}{\|u_1\|} \cdot u_1 = \frac{1}{\sqrt{2}} \cdot u_1 = \begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix}, \quad \|w_1\| = 1.$$

$$w_2 = \frac{1}{\|u_2\|} \cdot u_2 = \frac{1}{2\sqrt{3}} \cdot u_2 = \begin{bmatrix}\frac{3}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\\\frac{1}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\end{bmatrix}, \quad \|w_2\| = 1$$

$\therefore$ an orthonormal basis for V is $\left\{\begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix}, \begin{bmatrix}\frac{3}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\\\frac{1}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\end{bmatrix}\right\}$

0.5/5

Problem 8. [5] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

~~If n is odd~~.

A is an nxn square matrix satisfying $A^T = -A$.

If n is ~~odd~~ odd, then the diagonal entries of A should all be zero

~~then~~ (i.e. A should be $\begin{bmatrix} 0 & & \\ & \ddots & \\ & & 0 \end{bmatrix}_{n\times n}$ (the diagonal entries should all be zero).

(n=2m+1, m≥0.)

~~∴ A matrix A~~

~~In this case, A is not invertible~~

~~When A is not invertible,~~

~~det A = 0.~~

$\because I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix}_{n\times n}$ $\neq 0$.

Therefore, A cannot be written as a product of elementary matrices.

However, according to the theorem, Let A be an nxn matrix,

A is invertible if and only if it can be written as a product of elementary matrices.

∴ A is not invertible.

When A is not invertible, det A = 0.

5/5

Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

$\because |\vec{u}+\vec{v}| = |\vec{u}-\vec{v}|$

$\therefore |\vec{u}+\vec{v}|^2 = |\vec{u}-\vec{v}|^2$

$\therefore (\vec{u}+\vec{v})\cdot(\vec{u}+\vec{v}) = (\vec{u}-\vec{v})\cdot(\vec{u}-\vec{v})$

$\therefore |\vec{u}|^2 + \vec{v}\cdot\vec{u} + \vec{u}\cdot\vec{v} + |\vec{v}|^2 = |\vec{u}|^2 - \vec{v}\cdot\vec{u} - \vec{u}\cdot\vec{v} + |\vec{v}|^2$

$\because \vec{u}\cdot\vec{v} = \vec{v}\cdot\vec{u}$

$\therefore 4\vec{u}\cdot\vec{v} = 0$

$\therefore \vec{u}\cdot\vec{v} = 0$

$\therefore \vec{u}$ and $\vec{v}$ are perpendicular to each other.